

recommendations, with the only exception being one standard plan showing minor deviation in CI using the RTOG formula.

The minimum dose to PTV remained similar regardless of prescription method, however the D99% increased (range 2-6Gy) with reduction in pSI.

Reduction in pSI also showed a clear increase in monitor units, D2% and maximum point dose; up to 131% in the 80% plans and up to 166% in the 70% plans. The HI also increased accordingly.

In general terms GI reduced with reduction in pSI. Interestingly, all plans for three patients with PTVs of >65cm³ met the GI recommendation of <3.0 and none of the plans for two patients with PTVs <14cm³ did so.

In 7 out of the 10 cases, the organ at risk (OAR) doses were equivalent or lower in both the 70% and 80% pSI plans compared to the standard plans. For the three remaining cases, review of the standard plans suggests this is reflection of the optimisation methods rather than poorer dose gradient.

Conclusions: For hypofractionated treatments, where dose escalation within the target is highly desirable whilst keeping the OAR doses as low as possible, our results indicate that prescribing at the 70% pSI is preferable in most cases. However the results are influenced by the PTV size, optimisation method and vicinity of OARs.

References

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Stereotactic ablative body radiotherapy (SABR) for NSCLC oligometastatic patients: feasibility and outcomes

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Purpose/Objective: Data in the literature suggest the existence of oligometastatic disease, a state in which metastases are limited in number and site. Different kinds of local therapies have been used for the treatment of limited metastases and in the recent years reports on the use of Stereotactic Ablative radiotherapy (SABR) are emerging and the early results on local control are promising.

Aim: to evaluate local control, toxicity and overall survival in NSCLC oligometastatic patients underwent SABR.

Materials and Methods: Between October 2010 and May 2013, 36 NSCLC patients for 58 lung lesions were treated at our Institution. SABR was performed in case of controlled primary tumor, long interval time from the first diagnoses (greater than 6-12 months), exclusion of surgery, and number of metastatic sites ≤ 5. The total dose prescribed varied according to tumor site and maximum diameter. Dose prescription was 48 Gy in 4 fractions for peripheral lesions, 60 Gy in 8 fractions for central lesions and 60 Gy in 3 fractions for peripheral lesions with diameter ≤ 2 cm.

Results: Radiological response was obtained in the vast majority of patients. The local control at 1, 2 and 3 years was 100%, 95% and 95% respectively. No pulmonary toxicity G3-G4, chest pain or rib fracture occurred. The median follow up was 38 months (range 6-57 months). Overall Survival (OS) at 1,2 and 3 years was 90%,84% and 82% respectively.

Conclusions: SABR is feasible with limited morbidity. We believe that the discussion within a multidisciplinary team is of pivotal importance to select patients with better prognosis.

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CyberKnife® Re-irradiation For Recurrent Low-Grade Glioma

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Purpose/Objective: To evaluate the efficacy and toxicity of CyberKnife® fractionated stereotactic radiotherapy (FSRT) performed as re-irradiation in patients with recurrent low-grade glioma (LGG).

Materials and Methods: Ten patients were treated with CyberKnife® FSRT for recurrent LGGs re-irradiation in a single institution. At primary diagnosis, the histology of the tumor was grade II astrocytoma or oligodendroglioma. All recurrences were in-field or localized at the field border. Four out of 10 patients underwent surgery as primary treatment of recurrence. No histological upgrading was detected except one patient with radiological proof of high-grade differentiation.

Results: All patients underwent CyberKnife® FSRT with skull-base tracking method. No concurrent chemotherapy was applied except one with radiological tumour upgrading. Median prescription dose was 24 Gy (range 21-25 Gy) with a median fractionation number of 4 (range, 3-5 fractions). Median prescription isodose was 91% (range 80-94 %). Median value for PTV V95 and V107 was 100% and 38%, respectively. Brainstem and optic pathway maximum dose median value was 5 (range 2-27 Gy) and 2 (range 1-13 Gy), respectively.

CyberKnife® FSRT was well tolerated by all patients. No acute or late side effects recorded. Median overall survival from the initial diagnosis to the time of analyses was 117 months (range 47-213 months). Median interval between the first diagnosis and recurrence was 95 months (range 45-191 months). From the time point of re-irradiation, median survival was 15 months (range 1-30 months). Three out of 10 patients died because of comorbidities (n=1) and tumor progression (n=2). With a median follow-up of 17 months in patients who are alive, tumor progression is detected in 2. Five out of 10 patients are alive with local tumor control and improvement in performance status.

Conclusions: FSRT is well tolerated and showed encouraging short term results in this retrospective analysis of limited number of cases. Prospective studies are warranted for further evaluation.

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Fractionated stereotactic radiation with CyberKnife® for uveal melanoma : early results.

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Purpose/Objective: Radiation therapy is the gold-standard for the treatment of uveal melanoma. The technique of choice for small and median-sized melanomas is charged-ion radiation therapy (Proton, Helium). I¹²⁵ Brachytherapy may be considered for small tumours, but access to these two methods of treatment may be an issue, especially for elderly or disabled patients.

We present here the clinical results of 16 patients treated with fractionated stereotactic radiation with the Cyberknife® system in an entirely non invasive setting.

We report the control and toxicity rates for this treatment at 30 months of median follow-up.

Materials and Methods: We report a retrospective serie of 16 patients treated for uveal melanoma between may 2008 and may 2013, that had refused or were unsuitable for therapeutic alternatives (Proton-ion therapy, brachytherapy, enucleation). Patients were treated with 45 Gy in 3 fractions of 15 Gy.

Ocular immobilisation was obtained by palpebral occlusion according to Miralbell et al. (Tcrt 2007). The analysis of ocular displacements and subsequent margins is reported elsewhere.

Every patient underwent repeated ophthalmologic examination, and local control was assessed with ocular US scan and/or MRI.

We report the local and metastatic control rates, the ocular preservation rate, and toxicities.

Results: Median follow-up is of 30 months (3-58 months). Median age at treatment time was 70 years. Most tumours were of small or median size. The mean diameter was 9.4 mm, while the mean width was 4.8 mm.

In matter of tumour control, one local relapse was debated, but not alleged, as that patient was also diagnosed with metastases and was lost to follow-up. The local control rate was thus of 94-100%. Two metastatic relapses were alleged to the liver at 19 and 21 months, that is a metastatic control rate of 88%. Finally, 2 enucleations were indicated, both after treatment for irideal/ciliary body tumours, of which one was refused by the patient.

Grade 3-4 radiation complications were observed in 8 out of 16 patients (50%): 4 retinal detachments, 1 cataract, 3 radiation retinopathy, 1 intravitreous haemorrhage and 2 neovascular glaucomas. Treatments were as follow: vitrectomy and phakoemulsification (1 patient), intravitreous bevacizumab injections (6 patients, median of 4 injections), panphotocoagulation (5 patients), and laser iridotomy /IV hypotonic treatment (1 patient). Treatment recourse classified the